Measles

RQ: How does increasing measles vaccination coverage in the DRC reduce the amount of measles cases in the 2019 measles outbreak?

Key:

1. State variables:
   - S = Susceptible
   - I = Infectious
   - R = Recovered
   - V = Vaccinated
2. Parameters:
   - Λ = Births
   - $\lambda = \beta * I/N$
   - γ = Recovery rate
   - α = Disease induced mortality
   - μ = Background mortality
   - θ = Vaccination rate

$N = S + I + R + V$

<u>Eqns:</u>

- $dS/dt = \Lambda - \lambda S – VtS$
- $dI/dt = \lambda S - \mu I - \alpha I - \gamma I$
- $dR/dt = \gamma I - \mu R$
- $dV/dt= \theta S - \mu V$

<u>Model description</u>

Susceptible people (S) enter the population at per-capita birth rate Λ. Susceptible people can be infected through contact with an infectious person and experience a force of infection λ proportional to the prevalence of infectious people in the population ($\lambda = \beta * I/N$). Infected people enter the infectious class (I), where they can transmit the virus. The infectious people can recover at a rate γ and move to the recovered class where they will have lifelong immunity against the virus. All people experience background mortality at a per- capita hazard μ irrespective of infectious status, and infectious people experience an additional disease induced hazard or mortality α. Susceptible people are vaccinated rate θ and move into the vaccinated class (V), where they are protected from infection, the vaccinated class still experiences background mortality.

<u>Assumptions:</u>

- The population is not closed; births occur at a rate Λ and deaths occur at a rate μ.
- The population mixes homogeneously, meaning every individual has an equal chance of contacting every other individual.
- contact between a susceptible and infectious individual can result in infection.
- The force of infection is proportional to the prevalence of infection: $\lambda = \beta * I/N$.
- Recovered individuals acquire lifelong immunity and cannot be reinfected.
- Vaccinated individuals acquire lifelong immunity and cannot be Infected.
- Vaccination is administered to only susceptible people at a rate θ.
- All individuals receive background mortality rate μ
- Disease induced mortality occurs only in infectious individuals at a rate α.
- The transmission rate β, recovery rate γ, disease induced mortality rate α, and background mortality rate μ remain constant during the outbreak.
- The vaccination scenarios considered are 70%, 80%, 90%
- Once recovered or vaccinated you are immune from getting the virus
- Percentage of people vaccinated: 70%, 80% and 90% coverage.
- There is no migration into or out of the population.
- N = S + I + R + V